

HindsightTag: A Synaptic Tagging-and-Capture Framework for Retroactive Memory Consolidation in LLM Agents

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Abstract

Memory systems for LLM agents typically fix a memory’s importance at write time and adjust it only through later access to that same memory — a pattern we call the *encoding-time-monotonic salience assumption*. Human memory does not work this way: synaptic tagging-and-capture [Frey and Morris, 1997] and behavioral tagging [Moncada and Viola, 2007] show that a weak, otherwise-forgettable experience can be stabilized into long-term memory by an unrelated, sufficiently salient event occurring nearby in time. We formalize this as **HindsightTag**, a discrete, auditable mechanism with two parts: a backward rescue operator that retroactively promotes decaying memories via short-lived synaptic tags and a bounded capture window, and a temporal co-allocation mechanism that links memories written close in time independent of content. Both attach to an existing memory system’s lifecycle hooks without modifying its default behavior. Among the published LLM-agent memory systems we reviewed, we did not find this specific mechanism implemented — including in the two closest candidates, one of which names it only as unimplemented future work [Bering, 2026] and one of which explicitly restricts tagging to be non-retrospective [Lerma-Torres, 2026] — though our search was limited to published literature and did not cover patents or product documentation. We provide a reference implementation, a benchmark protocol (the Hindsight Rescue Benchmark), and preliminary results on real conversational data (LoCoMo, LongMemEval) supporting the mechanism’s non-interference property and a window-bounded rescue effect at short delays; these are an initial, limited-scale validation, not a mature or comparative evaluation, with full scope and caveats given in Section 5.

Keywords: LLM agent memory, synaptic tagging and capture, behavioral tagging, memory consolidation, retroactive salience, engram allocation, neuroscience-grounded AI architectures

1 Introduction

A large language model deployed as a long-running autonomous agent must decide, continuously, what to keep. The dominant engineering answer over the last three years has been retrieval: store everything as embeddings or graph triples, and rely on a similarity or graph-traversal function at query time to surface what matters [Lewis et al., 2020, Edge et al., 2024, Gutiérrez et al., 2024]. A newer and more demanding line of work has begun asking a different question — not “how do we find the right memory,” but “how does the system decide, on its own, what to remember, what to forget, and how to keep its knowledge from becoming stale, duplicated, or contradictory” [Du et al., 2025]. Systems answering this second question have increasingly borrowed directly from neuroscience: Ebbinghaus-curve decay [Wei et al., 2026, Gu et al., 2026], hippocampal indexing [Gutiérrez et al., 2024, 2025], complementary learning systems theory [McClelland et al., 1995, Lerma-Torres, 2026], and, most recently, explicit neuromodulator-inspired channels for dopamine, norepinephrine, serotonin, and acetylcholine [Bering, 2026]. We observe that every one of these systems — including the most sophisticated ones available at the time of writing

— shares a single, unexamined structural assumption. We call it the *encoding-time-monotonic salience assumption*: a memory’s importance score, decay rate, or retention priority is a function only of (a) the memory’s own content at the time it was written, and (b) the history of subsequent access to that same memory. Formally, for a memory m_i written at time τ_i , every system we are aware of computes importance as $I_i(t) = f(\text{content}(m_i), \{\text{access events on } m_i \text{ in } [\tau_i, t]\})$ for some system-specific function f . In the systems we examined, no term in the published f depends on events that happened to *other*, unrelated memories after τ_i .

This assumption is false of human memory, and the falseness is not an obscure edge case — it is a well-replicated effect within the synaptic tagging and behavioral tagging literature specifically. The synaptic tagging-and-capture (STC) hypothesis [Frey and Morris, 1997, Redondo and Morris, 2011] holds that a weak learning event leaves a short-lived molecular “tag” at the relevant synapses without being strong enough, on its own, to produce a stable long-term memory. If a second, sufficiently salient event occurs within a bounded time window — on the order of hours — the plasticity-related proteins it triggers are captured by the still-active tag from the first event, and the original weak memory is retroactively stabilized into a lasting one. Behavioral tagging [Moncada and Viola, 2007] extends this to whole-organism learning: a training session too weak to produce long-term memory on its own can be rescued into long-term memory by mere exposure to an unrelated novel environment occurring nearby in time. A large and still-active body of work continues to refine the temporal and associative boundaries of this effect; a 2025 study demonstrated that emotionally salient experiences can retroactively enhance recall of mundane experiences that preceded or followed them by a graded, non-binary amount [Lin et al., 2025], and other recent work has extended the known temporal window for tag-capture interaction considerably beyond initial estimates. Separately, a distinct but related body of research on engram allocation shows that neurons recruited to represent a new memory are selected partly by a transient excitability state, with the consequence that two unrelated experiences occurring within a few hours of each other come to share overlapping neural representations and become associatively linked — while the same two experiences separated by a day or more do not [Yiu et al., 2014, Rashid et al., 2016, Josselyn and Frankland, 2024]. Both phenomena describe something we did not find implemented in any of the AI memory systems examined in this study: the significance and connectivity of a memory can be determined, in part, by what happens *after* it, not only by what happens *to* it.

We verified two of the closest candidate systems directly rather than relying on citation review alone. The single closest prior system, ZenBrain [Bering, 2026], names the mechanism explicitly. In an appendix cataloguing capabilities beyond its fifteen evaluated algorithms, it states that a “Synaptic Tagging and Capture (STC) rescue module implementing Frey and Morris [1997]’s plasticity-donation paradigm. . . rescues fading memories when a nearby strongly-consolidated memory is activated within the STC window,” and immediately notes that this and three other capabilities “are omitted from the present evaluation for space.” No formula, parameter value, temporal window duration, pseudocode, or evaluation accompanies this sentence anywhere in the 47-page document, and it is not one of the paper’s fifteen ablated mechanisms. ZenBrain’s actual implemented reconsolidation mechanism [Bering, 2026] is, by contrast, purely retrieval-triggered: a memory becomes labile only when it is accessed, opens a ten-minute editable window, and is updated only in relation to new content presented for that same memory at that moment. Nothing in the implemented system reaches backward from a later, unrelated event to an earlier, non-retrieved memory. A second neuroscience-grounded architecture, the theoretical framework of Lerma-Torres [2026], states its position even more explicitly in the opposite direction: emotional tagging in that architecture “operates strictly on current working memory contents. . . not retrospectively.”

Beyond these two directly quoted systems, our claim is narrower. Across the broader set of published LLM agent memory systems we reviewed — including the multi-agent, six-memory-type architecture of Wang

and Chen [2025], discussed specifically in Section 2.1, and the forgetting- and decay-based systems of Section 2.3 — we did not find a mechanism in which a later, content-unrelated event changes the retention trajectory of an earlier, already-written memory. We did not verify this by direct source inspection for every system in that set the way we did for ZenBrain and Lerma-Torres above, and we did not search patents, internal product documentation, or unpublished work; commercial systems in particular (Section 2.1) may implement undocumented behavior that differs from their published technical reports. We state this scope explicitly rather than implying a broader search than we performed. We also note that a structurally related but distinct research tradition — temporal-locality and working-set-style grouping in caching and information retrieval, where recent proximity to *any* access drives promotion or prefetching decisions [Denning, 1968] — addresses an adjacent problem (retrieval or cache efficiency) rather than the retroactive-salience-revision problem addressed here; we return to this adjacency where it bears most directly, on the temporal co-allocation mechanism, in Section 4.5. Within this scope, we separate four distinct kinds of contribution rather than treating “novelty” as a single claim: a *mechanism* contribution (a rescue operator triggered by an external, later event rather than by querying the memory itself); a *formalization* contribution (specifying synaptic tagging-and-capture and behavioral tagging as an algorithm over discrete, timestamped, auditable symbolic records, where we did not find a comparable discrete formalization in the literature we searched); a *benchmark* contribution (an evaluation protocol targeting retroactive rescue behavior specifically); and an *implementation* contribution (a reference codebase and preliminary empirical results) — explicitly the weakest-evidence of the four, being a single implementation evaluated at limited scale. The list below is organized under these four categories. This paper makes the following contributions:

1. *Naming/formalization.* We identify and formally name the encoding-time-monotonic salience assumption as an implicit, previously unstated property shared by the surveyed literature on LLM agent memory, and give a general, falsifiable test for whether a system satisfies it (Section 3).
2. *Mechanism.* We specify a backward rescue operator whose triggering condition is a later, independent event rather than access to the memory being rescued — a trigger structure that, to our knowledge, differs from the retrieval-triggered reconsolidation in ZenBrain and the per-query retrieval reweighting in SynapticRAG, in that it produces a permanent, audited state change to a memory independent of whether or when that memory is subsequently queried (Section 4).
3. *Formalization.* To the best of our knowledge, we give the first specification of synaptic tagging-and-capture and behavioral tagging as an algorithm operating over discrete, timestamped, auditable symbolic memory records with an explicit host-interface contract, as distinct from continuous-weight computational treatments of tagging in connectionist modeling, where they exist (Section 4).
4. *Companion mechanism (narrower novelty claim).* We formalize temporal co-allocation, a content-independent temporal-proximity linking mechanism. We claim novelty only in its specific formalization — a symmetric, weighted graph edge integrated with the rescue mechanism above, with an arbitrary, non-session-bounded lookback window — not in the general idea of temporal grouping, which is well established in the working-set/temporal-locality caching tradition (Section 4.5).
5. *Engineering.* We show both mechanisms can be attached to an existing memory architecture’s life-cycle hooks as a non-interfering plug-in — confirmed exactly, not just by design, on a non-decaying host (Section 5.4) — and provide concrete implementation guidance and a complexity analysis for doing so (Sections 4.6–4.8).
6. *Benchmark.* We specify a complete evaluation protocol, the Hindsight Rescue Benchmark, designed specifically to measure retroactive rescue behavior, which no existing benchmark [Maharana et al., 2024, Wu et al., 2024, Hu et al., 2025] appears to be constructed to test (Section 5).

7. *Implementation (weakest-evidence contribution, labeled as such).* We implement the mechanism in a reference codebase and report preliminary empirical results on real conversational data (LoCoMo and LongMemEval), presented as an initial, limited-scale validation that the mechanism behaves as specified, not a comparative or production-scale evaluation (Section 5).

2 Related Work

2.1 Retrieval-centric memory for LLM agents

The dominant paradigm for giving an LLM agent persistent memory is retrieval-augmented generation: embed past content, store it in a vector index, and retrieve the top- k most similar items at query time [Lewis et al., 2020]. Knowledge-graph extensions replace or augment this with explicit entity-relation extraction. HippoRAG [Gutiérrez et al., 2024] constructs an open knowledge graph from LLM-extracted triples and performs associative multi-hop retrieval via Personalized PageRank, explicitly modeling the division of labor between hippocampal indexing and neocortical storage; its successor [Gutiérrez et al., 2025] improves associative and sense-making retrieval further. Microsoft’s GraphRAG [Edge et al., 2024] instead builds a hierarchy of community summaries over an extracted knowledge graph to answer corpus-wide sensemaking questions that no single retrieved passage could answer alone. Zep [Rasmussen et al., 2025] stores conversational and business context as a bi-temporal property graph, allowing edges to be time-scoped and invalidated as new information supersedes old. Supermemory [Daga et al., 2026] similarly targets long-term conversational memory, reporting state-of-the-art results on LongMemEval-S through chunk-based ingestion into disambiguated atomic memories, a relational versioning scheme (updates, extends, and derives edges) that tracks how facts are corrected or refined as new information arrives, and dual-timestamp temporal grounding (a `documentDate` versus an extracted `eventDate`) that supports reasoning about when events occurred relative to when they were reported. Its updates/extends/derives edges, like Mem0’s and Zep’s mechanisms below, are formed between an arriving memory and an existing one on the basis of content relatedness; the technical report we reviewed does not describe a decay or forgetting process for its memories, and none of its five described components lets a later, content-unrelated event change the retention of an earlier, low-salience memory. We note this claim reflects the published report as of our review and not necessarily the current deployed product, which may have since changed given the pace of development in this space. The systems reviewed in this subsection are architecturally organized around improving *what is retrieved*; none of them defines a mechanism by which a later event changes the standing of an earlier, unrelated memory.

MIRIX. MIRIX [Wang and Chen, 2025] is a multi-agent memory system organized around six memory types (Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault), coordinated by a central Meta Memory Manager and per-type Memory Managers, and evaluated on LOCOMO and a multi-modal screenshot-QA benchmark. Its memory-update pathway is entirely content-routed: on new input, the Meta Memory Manager “analyzes the content and determines which memory components are relevant, routing the input to the corresponding Memory Managers,” which “update their respective memories in parallel while ensuring that redundant information is avoided within each memory type.” The paper’s own worked example of retroactive-seeming behavior is instructive: MIRIX stores the consolidated event that a person went camping on a specific date, and this later, confirmed occurrence takes precedence over an earlier stated plan to go camping, because both statements refer to the *same* camping event. This is a content-relatedness-gated consolidation of the kind already discussed for FadeMem and Mem0 above, not a case of a later, unrelated event changing the fate of an earlier, dissimilar, low-salience memory. We did not find a decay, forgetting, or salience-based tagging mechanism described anywhere in the paper; nothing in the described

system appears to allow a memory to fade and later be retroactively rescued, since there is no described process by which memories fade in the first place.

2.2 Synaptic- and temporally-inspired retrieval mechanisms

Closest in spirit to the present paper is SynapticRAG [Hou et al., 2024], which also borrows synaptic terminology to address temporal structure in LLM memory retrieval, and which we address directly given the overlap in framing. SynapticRAG scores each stored dialogue turn with a continuous, biologically-inspired propagation signal — a dynamic leaky integrate-and-fire mechanism operating over temporal association triggers — so that a query’s activation spreads to temporally and associatively nearby turns and shifts their retrieval ranking; the mechanism reports consistent gains over similarity-only retrieval across English, Chinese, and Japanese conversational benchmarks. This is a real, evaluated system, and it establishes that synaptic metaphors for temporal memory in LLM agents are not unprecedented. It differs from HindsightTag along two structural axes rather than by degree: first, SynapticRAG’s propagation signal continuously reweights *retrieval ranking* at query time, whereas HindsightTag’s tag-decay-and-capture cycle (Section 4) discretely and permanently changes a memory’s own *retention state* $v_i(t)$ in the host store, independent of whether or when it is later queried; second, HindsightTag’s rescue transition (Equation 5) is triggered by a specific, identifiable later event crossing an explicit salience threshold and produces a permanent, audited provenance record (Section 4.8), whereas SynapticRAG’s propagation is a continuous, per-query scoring function with no discrete triggering event or audit trail. To our knowledge, no published system — including SynapticRAG — implements a discrete, threshold-gated, auditable backward rescue operator of the form specified in Section 4; synaptic-inspired mechanisms for LLM memory are not themselves unprecedented, and our novelty claim is scoped to this specific mechanism rather than to the broader use of synaptic metaphors.

2.3 Forgetting- and decay-based memory architectures

A second line of work treats forgetting itself as a first-class design problem. FadeMem [Wei et al., 2026] assigns each memory a dual-layer decay profile governed by an Ebbinghaus-style exponential curve, modulated by access frequency and semantic relevance, and performs LLM-adjudicated conflict resolution and fusion of related memories. FSFM [Gu et al., 2026] proposes a taxonomy of passive decay-based, active deletion-based, safety-triggered, and adaptive reinforcement-based forgetting, combined into a single importance score used for capacity-constrained pruning, and validates the approach on production telecommunications-assistant data. Mem0 [Chhikara et al., 2025] performs per-turn, LLM-adjudicated ADD, UPDATE, and DELETE operations against retrieved similar memories. In all three systems, cross-memory interaction, where it exists, is gated by content similarity to a newly arriving item: FadeMem’s conflict-resolution module compares a new memory only against existing memories retrieved above a similarity threshold, and Mem0’s ADD/UPDATE/DELETE decision is likewise made only against top- k retrieved similar memories. In both cases the resulting update suppresses, merges, or ages the *older, similar* memory; we did not find, in either system, a channel by which the arrival of a later, content-*unrelated* event retroactively promotes an earlier, dissimilar, low-salience memory that would otherwise decay away.

2.4 Neuroscience-grounded agent memory architectures

Two very recent architectures adopt an explicitly neuroscience-first design philosophy, and are the closest prior work to the present paper. ZenBrain [Bering, 2026] implements a seven-layer memory architecture (working, short-term, episodic, semantic, procedural, core, and cross-context memory) coordinated by fifteen validated mechanisms, including a two-factor Hebbian synaptic model for knowledge-

graph edges, a prediction-error-gated reconsolidation engine, a triple-copy decay model with fast, medium, and logarithmic-growth deep components, and a four-channel neuromodulator engine explicitly modeling dopamine, norepinephrine, serotonin, and acetylcholine as distinct computational signals rather than a single collapsed importance score. It reports strong empirical results against Mem0, Letta, and A-Mem on LongMemEval and LoCoMo. As detailed in Section 1, it references synaptic tagging and capture once, as an unimplemented, unevaluated appendix note.

Human-Like Lifelong Memory [Lerma-Torres, 2026] is a purely theoretical architecture proposal, explicitly following the tradition of prior computational-level theorizing in memory research [McClelland et al., 1995] rather than presenting an implementation. It proposes a thalamic gateway that tags incoming information along six salience channels (thematic relevance, emotional charge, urgency, novelty, source trust, and goal affinity), a knowledge-graph memory service organized around compact valence vectors, and a System 1/System 2 retrieval architecture modeled on dual-process cognition [Kahneman, 2011]. Its gist-formation mechanism is explicitly restricted to present-moment tagging, and its one discussion of retroactive improvement concerns gradual, graph-density-driven retrieval quality across the whole memory store over time, not a discrete event retroactively rescuing one specific prior trace.

2.5 Neuroscience foundations: synaptic tagging, behavioral tagging, and engram allocation

The synaptic tagging-and-capture hypothesis was first proposed by Frey and Morris [1997], who showed that a weak stimulation protocol insufficient to produce lasting long-term potentiation on its own could nonetheless be stabilized into a lasting change if a strong, protein-synthesis-dependent stimulation occurred at a different synaptic input on the same neuron within a bounded time window. Redondo and Morris [2011] provide a comprehensive review of the hypothesis and its behavioral implications. Moncada and Viola [2007] demonstrated the behavioral analogue directly: a weak inhibitory-avoidance training session, insufficient to produce long-term memory on its own, was rescued into long-term memory by exposure to an unrelated novel open field occurring close in time to training — establishing that the rescuing event need not be semantically or associatively related to the memory it rescues. A substantial subsequent literature has refined the temporal window and boundary conditions of this effect; recent work demonstrates that the effect persists over considerably longer intervals than initially estimated, and a 2025 study in *Science Advances* shows that the retroactive enhancement of mundane, co-occurring memories by a salient event is graded rather than all-or-none, and operates both backward and forward in time relative to the salient event [Lin et al., 2025]. We are not aware of any prior work that formalizes this specific phenomenon as an algorithm operating over discrete symbolic memory records for an artificial agent. Where computational treatments of synaptic tagging and capture exist in the connectionist-modeling literature, they are, to our knowledge, framed at the level of continuous synaptic weight dynamics rather than discrete, auditable records; we did not conduct an exhaustive review of that literature, since it targets a different representational level and a different problem (neural weight dynamics versus auditable knowledge-record management) from the one addressed here, and we do not cite specific such models as we cannot verify individual instances with confidence.

A related and largely independent literature concerns engram allocation and memory linking. Neurons are not recruited to a new memory trace at random; those with transiently elevated excitability — driven in part by the transcription factor CREB — are preferentially allocated [Yiu et al., 2014]. Because this excitability window is itself time-bounded, two memories encoded within a few hours of one another come to share overlapping neural populations and become linked, such that later reactivation of one engram can influence recall of the other, whereas memories separated by a day or more do not share this overlap and remain functionally independent [Rashid et al., 2016]. Josselyn and Frankland [2024] synthesize this line of work into a general account of engram-based memory linking and identity formation. As with synaptic tagging

and capture, this phenomenon is defined by temporal proximity of encoding rather than by content similarity — the opposite organizing principle from the semantic-embedding- and knowledge-graph-based indexing schemes in the LLM agent memory systems surveyed in Sections 1–2.3. (A non-neuroscience research tradition organized around temporal proximity in a different sense — locality of reference in program and cache behavior — is discussed in Section 4.5.)

2.6 Positioning of this work

The systems reviewed above collectively cover salience scoring, decay, consolidation, contradiction resolution, temporal validity, and even explicit neuromodulator-style channels. Among the systems reviewed above — including the two most directly comparable and most recently published neuroscience-grounded architectures — we did not find one that implements a mechanism by which a later, temporally subsequent event changes the retention trajectory of an earlier, non-retrieved, unrelated memory, nor one that implements content-independent temporal-proximity binding as a retrieval mechanism distinct from semantic similarity. HindsightTag is designed to fill precisely this gap, as a plug-in addition compatible with, rather than a replacement for, the systems reviewed here. We note for disambiguation that an unrelated, concurrently published system, *Hindsight is 20/20* [Latimer et al., 2025], shares the word “Hindsight” in its name and targets the same benchmarks (LongMemEval, LoCoMo) but addresses a different problem: it organizes agent memory into four logical networks (world facts, agent experiences, entity summaries, evolving beliefs) governed by retain/recall/reflect operations, aimed at retrieval accuracy and explainable reasoning traces, and does not implement a retroactive rescue mechanism of the kind specified in Section 4. The two systems are unrelated in mechanism despite the naming overlap.

3 Problem Formulation: The Encoding-Time-Monotonic Salience Assumption

We define the assumption precisely so that it can be falsified by future systems. Let $M = \{m_1, m_2, \dots, m_N\}$ denote a memory store, where each m_i is written at time τ_i and carries a retention or strength trajectory $v_i(t) \geq 0$ defined for $t \geq \tau_i$, governed by some host system’s decay and reinforcement rules. We say a memory system satisfies encoding-time-monotonic salience if, for every m_i and every $t > \tau_i$,

$$v_i(t) = D(v_i(\tau_i), \{a : a \text{ is an access or reinforcement event on } m_i \text{ in } [\tau_i, t]\}; \theta) \quad (1)$$

for some decay/reinforcement function D and parameter set θ that depends only on m_i ’s own history — that is, $v_i(t)$ is conditionally independent of the content, arrival, or salience of any other memory m_j ($j \neq i$) written after τ_i .

All systems examined in this study satisfy this property in the strict sense above, with one qualification we make precise here. A partial exception arises in systems that perform similarity-gated conflict resolution or fusion (Section 2.3): there, $v_i(t)$ can indeed depend on a later memory m_j , but, as detailed above, only when a content-relatedness term between m_i and m_j exceeds the host system’s own similarity threshold, and only to suppress, merge, or age m_i . We therefore state the operative form of the assumption used for the remainder of this paper: in all systems we examined, cross-memory dependence, where present, is (i) restricted to content-similar pairs, and (ii) monotonically non-increasing in its effect on the older memory’s retention. Ebbinghaus-curve decay [Wei et al., 2026, Gu et al., 2026] is manifestly a function of m_i ’s own elapsed time and access count. ZenBrain’s triple-copy decay and two-factor Hebbian consolidation [Bering, 2026] update edge weights and variances based on that edge’s own co-activation history. Reconsolidation in both ZenBrain and the reconsolidation literature it draws on [Nader et al., 2000] is triggered by retrieval of

the memory itself, and updates it only in relation to new content offered for that same memory at that moment — the update is conditioned on m_i being accessed, not on an unrelated m_j occurring later. HindsightTag (Section 4) departs from this operative assumption along both axes at once: capture can occur without content relatedness ($\omega_{\text{assoc}} = 0$; Section 4.4), and its effect on the rescued memory is retention-*increasing* rather than decreasing.

This assumption is a reasonable engineering default: it makes each memory’s lifecycle independently analyzable and avoids unbounded cross-memory interaction costs. But it is also the precise point at which the surveyed literature diverges from the biological system it otherwise draws extensively upon, and it forecloses a specific, useful class of behavior: the ability to recognize, after the fact, that an earlier, low-salience event was actually a precursor to something significant. Section 4 formalizes a mechanism that relaxes this assumption in a bounded, auditable, and computationally tractable way.

4 HindsightTag: Method

Figure 1 gives an overview of how the mechanisms specified in this section relate to one another and to a host memory system before we define them formally.

4.1 Preliminaries and notation

We assume a host memory system that already provides, for each memory m_i , a content representation, a timestamp τ_i , a retention/strength value $v_i(t) \in [0, 1]$, and some existing salience or importance function $\text{Sal}(\cdot) \in [0, 1]$ used to gate writes or drive decay (any of the systems in Sections 2.2–2.3 provide such a function). HindsightTag does not replace $v_i(t)$ or $\text{Sal}(\cdot)$; it adds a second, independent state variable per memory and a small set of triggered operators that act on $v_i(t)$ from the outside.

4.2 Synaptic tags and tag decay

At the moment a memory m_i is written, if its initial salience $\text{Sal}(m_i)$ falls below the host system’s own consolidation threshold (i.e., it would otherwise simply decay away), HindsightTag assigns it a synaptic tag $g_i(t)$, initialized to a value proportional to its initial encoding strength and decaying independently of, and considerably faster than, the memory’s own retention curve:

$$g_i(t) = g_0 \cdot \exp\left(-\frac{t - \tau_i}{T_{\text{tag}}}\right), \quad \tau_i \leq t \leq \tau_i + W_{\text{capture}}; \quad g_i(t) = 0 \text{ otherwise}, \quad (2)$$

where $g_0 = \max(v_i(\tau_i), \text{Sal}(m_i))$ — the tag is seeded from whichever of the memory’s initial retention or salience is larger, so that a memory the host under-weights on retention but that carries meaningful salience at write time is not denied a tag purely as an artifact of the host’s own initialization policy — and $T_{\text{tag}} \ll$ the host system’s typical retention half-life, reflecting the biological finding that a synaptic tag is a short-lived, minutes-to-hours phenomenon [Frey and Morris, 1997, Redondo and Morris, 2011] that exists independently of, and much more briefly than, the resulting long-term memory it may eventually enable. W_{capture} is a bounded lookback/lookahead window, a hyperparameter requiring empirical calibration (Section 7); recent evidence suggests this window may extend considerably beyond early estimates [Lin et al., 2025], and we deliberately leave its exact value unspecified pending the evaluation described in Section 5. Section 4.8 gives literature-informed starting ranges for T_{tag} and W_{capture} to seed calibration.

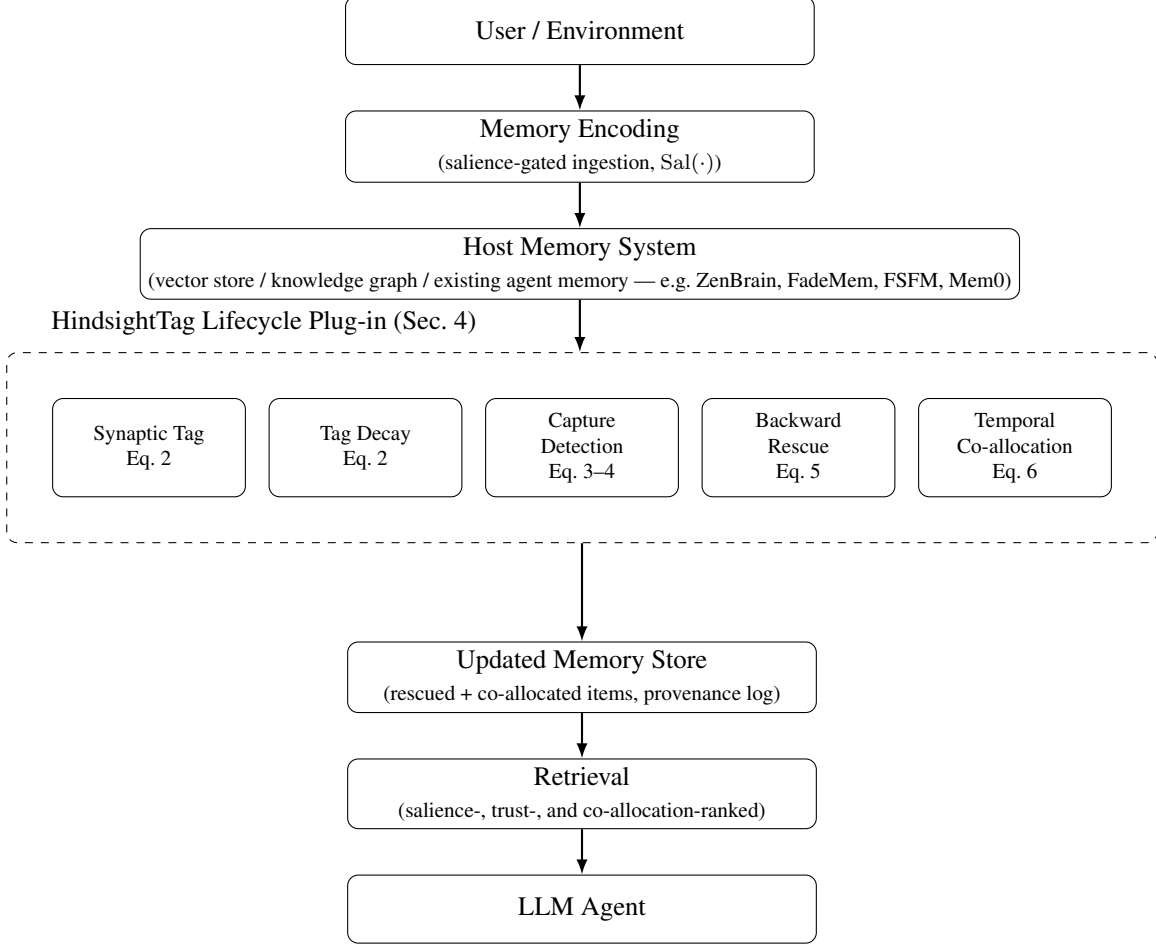


Figure 1: HindsightTag as a lifecycle plug-in attached to a host memory system. Incoming events are encoded and salience-gated by the host system as usual; the plug-in (dashed box) adds five bounded, non-interfering operations — synaptic tagging, tag decay, capture detection, backward rescue, and temporal co-allocation (Sections 4.2–4.5) — that act on the host system’s memory store before retrieval. No component of the host system (top) or the retrieval/agent loop (bottom) needs to be modified; the plug-in only adds operations at the `store()`, `decay()/tick()`, and `consolidate()` hooks described in Section 4.6.

4.3 Capture events and the backward rescue operator

Let e denote any newly arriving event with host-system salience $\text{Sal}(e)$. We say e triggers a *capture event* if $\text{Sal}(e) \geq \theta_{\text{capture}}$ for a threshold θ_{capture} set by the deployer. Upon a capture event at time t_e , HindsightTag scans the set of currently tagged, not-yet-consolidated memories whose tags remain active:

$$T_{\text{active}}(t_e) = \{ m_i : \tau_i \in [t_e - W_{\text{capture}}, t_e), g_i(t_e) > 0 \}. \quad (3)$$

For each $m_i \in T_{\text{active}}(t_e)$, we compute a capture strength

$$C_i = g_i(t_e) \cdot \text{Sal}(e) \cdot [\omega_{\text{assoc}} \cdot \text{Rel}(e, m_i) + (1 - \omega_{\text{assoc}})], \quad (4)$$

where $\text{Rel}(e, m_i) \in [0, 1]$ is an optional content-relatedness term (e.g., embedding cosine similarity) and $\omega_{\text{assoc}} \in [0, 1]$ is a mixing weight. If m_i is rescued via a capture event where $\text{Rel}(e, m_i)$ is deliberately excluded or $\omega_{\text{assoc}} = 0$, we call this *heterosynaptic capture*, directly reflecting the finding that behavioral

tagging succeeds even when the rescuing event is unrelated to the rescued memory [Moncada and Viola, 2007]. If $\omega_{\text{assoc}} = 1$, capture requires content relatedness and reduces to a purely associative variant. We treat ω_{assoc} as a deployment-time design choice rather than a fixed constant, since both regimes are independently supported by the neuroscience literature (Section 2.4) and their relative value in an artificial-agent setting is an open empirical question we defer to Section 5. If $C_i \geq \theta_{\text{rescue}}$, m_i undergoes a rescue transition:

$$v_i(t_e^+) = \min(1, v_i(t_e) + \Delta_{\text{rescue}} \cdot (1 - v_i(t_e)) \cdot C_i), \quad (5)$$

m_i 's status is updated from provisional/tagged to consolidated (or the host system's equivalent terminal retention state), and a provenance record — $\{\text{rescued_by} : e, \text{at} : t_e, \text{capture_strength} : C_i\}$ — is permanently appended to m_i 's audit log (a concrete schema is given in Section 4.8). If no capture event occurs before $g_i(t)$ reaches zero, m_i is left entirely untouched by HindsightTag and proceeds through the host system's ordinary decay pathway; this non-interference property is by design (Section 4.6).

4.4 Heterosynaptic versus associative capture modes

Because ω_{assoc} is exposed as a parameter rather than fixed, HindsightTag supports both of the two regimes documented in the neuroscience literature within a single formalism: pure heterosynaptic capture [Moncada and Viola, 2007], in which any sufficiently salient event can rescue any tagged memory within the window regardless of content, and associative capture, in which rescue additionally requires the incoming event to be contextually related to the tagged memory. We expect the appropriate regime to be domain-dependent — a purely heterosynaptic default risks false rescues in high-throughput environments (Section 7) — and treat the calibration of ω_{assoc} as an explicit target of the evaluation protocol in Section 5.

4.5 Temporal co-allocation: content-independent engram linking

Independently of the rescue mechanism above, HindsightTag maintains a separate temporal graph $\mathcal{T}$ over the memory store, motivated by engram-allocation and memory-linking findings [Yiu et al., 2014, Rashid et al., 2016, Josselyn and Frankland, 2024]. At write time, every memory m_i is assigned an eligibility window of duration E_{window} (a second, independent hyperparameter). Any two memories m_i, m_j written such that $|\tau_i - \tau_j| \leq E_{\text{window}}$ receive a symmetric, weighted co-allocation edge

$$L(i, j) = \exp\left(-\frac{|\tau_i - \tau_j|}{E_{\text{window}}}\right), \quad (6)$$

added to $\mathcal{T}$ regardless of the semantic or graph-based similarity between m_i and m_j as computed by the host system. Retrieval of m_i can then additionally surface its temporal co-allocates $\{m_j : L(i, j) \geq \theta_{\text{link}}\}$, giving the memory system a second retrieval axis — “what else was recorded around this time” — that is structurally distinct from, and additive to, the embedding- and knowledge-graph-similarity-based retrieval paths in the LLM agent memory systems reviewed in Section 2. We note that this mechanism is deliberately simpler than the full biological account of CREB-mediated excitability, which we do not attempt to model at the level of individual computational units; we adopt the functional consequence (temporal-proximity-driven, content-independent linking) rather than the cellular mechanism producing it.

We also note an adjacent, non-neuroscience research tradition that motivates a narrower framing of this contribution. The working-set model of program behavior [Denning, 1968] and the broader locality-of-reference literature it founded in operating systems and caching establish that recent temporal proximity to *any* access is a useful, content-independent signal for what to keep resident or prefetch; session-windowing heuristics in some deployed conversational memory systems apply a related idea informally, grouping items

by conversation or time-bucket boundaries. Temporal co-allocation shares the underlying intuition — proximity in time, independent of content, is informative — but targets a different problem (surfacing a specific unrelated memory as a retrieval result, via an explicit, arbitrarily-spanning, weighted graph edge integrated with the rescue mechanism of Sections 4.2–4.4) rather than cache residency or eviction efficiency. We claim novelty only in this specific formalization and its integration with backward rescue, not in the general idea of temporal-proximity-based grouping, and we have not yet benchmarked temporal co-allocation against a simple session- or time-bucket baseline drawn from this tradition (Section 7).

4.6 Integration as a lifecycle plug-in

HindsightTag is designed to attach to the small set of lifecycle operations that contemporary memory architectures already expose. Using ZenBrain’s `MemoryCoordinator` interface [Bering, 2026] as a concrete illustration of such an interface, HindsightTag requires only three additional hooks: on `store(item)`, initialize g_i and register m_i in the temporal graph $\mathcal{T}$, then immediately check whether the arriving `item` itself constitutes a capture event and, if so, execute the backward rescue operator (Equations 3–5) over T_{active} before returning; on the host system’s periodic `decay()/tick()` operation, update all active tags via Equation 2 and expire any tag whose window has closed; on `consolidate()`, tags that survive to natural consolidation are absorbed into the host system’s own consolidation pathway without modification, and expired tags are silently dropped. No existing host-system operation needs to be modified, and in the case where no capture event ever occurs for a given tagged memory, HindsightTag has zero effect on that memory’s trajectory — a deliberate design property intended to make the mechanism safe to attach to a production system incrementally. Section 4.8 specifies the minimal interface contract a host system must expose for this attachment to be well defined.

4.7 Complexity and scalability

The backward rescue operator (Equation 3) scans only $T_{\text{active}}(t_e)$, the set of memories tagged within the bounded window W_{capture} that have not yet expired or been consolidated. Because W_{capture} is bounded and tags are actively pruned once expired, $|T_{\text{active}}(t_e)|$ is bounded by the write rate of the host system times W_{capture} , independent of the total size N of the memory store. Each capture event therefore costs $O(k)$ rather than $O(N)$, where $k = |T_{\text{active}}(t_e)|$, and can be implemented with a time-indexed queue rather than a scan over the full memory store. The temporal co-allocation graph (Section 4.5) similarly requires only a bounded backward look at write time, using the same time-indexed structure, and does not require recomputing edges over the full store as new memories arrive.

4.8 Implementation guidance and reproducibility

Beyond the reference implementation used for the preliminary results in Section 5.4, we specify the mechanism at a level of detail intended to let an independent implementer reproduce it without further design decisions of their own, and to let future empirical studies be directly comparable across independent re-implementations.

Host-system interface contract. A host system attaching HindsightTag must expose, at minimum, the following four accessors, each already implicit in the systems reviewed in Section 2:

get_salience(item) → [0,1]

the host system’s own $\text{Sal}(\cdot)$ function, used both to decide whether a newly written item receives a tag (Section 4.2) and to detect capture events (Section 4.3);

get_timestamp(item) -> real

the write time τ_i , assumed monotonically non-decreasing in arrival order;

get_relatedness(item_a, item_b) -> [0,1]

an optional embedding- or graph-similarity function supplying $\text{Rel}(\cdot, \cdot)$ in Equation 4; if the host system omits this, HindsightTag must be run with $\omega_{\text{assoc}} = 0$ (pure heterosynaptic capture);

get_consolidation_threshold() -> [0,1]

the scalar below which the host system would otherwise let an item decay away untouched, used to decide tag eligibility at write time (Section 4.2).

No other host-system internals need to be inspected or modified; HindsightTag treats the host system as a black box behind this four-function interface plus the three lifecycle hooks of Section 4.6.

Recommended data structures. We recommend implementing T_{active} (Equation 3) as a min-heap keyed by tag-expiry time ($\tau_i + W_{\text{capture}}$), so that expired tags can be popped in amortized $O(\log k)$ time at each `tick()` rather than rescanned; and the temporal co-allocation graph $\mathcal{T}$ (Section 4.5) as a bucketed structure indexed by fixed-width time windows of size E_{window} , so that a new memory’s co-allocation edges (Equation 6) can be computed by inspecting only its own and adjacent buckets rather than the full store. Both structures make each of the operations in Sections 4.2–4.5 cost at most $O(\log k)$ or $O(k)$ in the number of currently tagged or temporally adjacent items, consistent with the complexity analysis of Section 4.7.

Provenance and audit log schema. We recommend the following minimal record format for every rescue event, to satisfy the auditability requirement discussed in Section 6.2:

```
{
  "memory_id":      <id of the rescued memory m_i>,
  "rescued_by":     <id of the triggering event e>,
  "at":             <timestamp t_e of the capture event>,
  "capture_strength": <C_i, Eq. 4>,
  "omega_assoc_used": <omega_assoc at time of capture>,
  "mode":           "heterosynaptic" | "associative" | "mixed"
}
```

and a parallel, permanent record for every item that is instead allowed to decay to a tombstoned state, recording at minimum `memory_id`, `last_tag_value`, and `expired_at`, so that a reviewer can distinguish “never tagged,” “tagged and expired unrescued,” and “tagged and rescued” after the fact.

Literature-informed starting ranges (not validated defaults). The five capture-mechanism hyperparameters (T_{tag} , W_{capture} , θ_{capture} , θ_{rescue} , ω_{assoc}) and the co-allocation window E_{window} have no validated values, since the preliminary run in Section 5.4 uses these literature-informed seeds rather than tuned values. To give an implementer a starting point for calibration rather than an unconstrained six-dimensional search, we suggest initializing T_{tag} on the order of the classical synaptic tagging window (on the order of one to a few hours, as originally characterized by Frey and Morris, 1997, Redondo and Morris, 2011), and W_{capture} at a somewhat longer duration, informed by evidence that the behaviorally relevant window can extend considerably further [Lin et al., 2025]. We emphasize that these are literature-informed starting points for the calibration procedure of Section 5, not recommended production values; θ_{capture} , θ_{rescue} , ω_{assoc} , and E_{window} have, to our knowledge, no directly analogous quantity reported in the neuroscience literature and should be treated as free parameters to be fit per deployment.

Reproducibility checklist. We recommend that any implementation or evaluation of HindsightTag: (1) log and version all six hyperparameters alongside every experimental run; (2) unit-test Equations 2–6 in isolation, against synthetic single-memory, single-event inputs, before integration-testing the full lifecycle of Section 4.6; (3) fix a deterministic tie-breaking rule (e.g., lowest `memory_id` first) for cases where multiple tagged memories reach identical capture strength C_i in the same capture event, so that outcomes are reproducible across runs and implementations; (4) fix random seeds for any stochastic host-system component (e.g., an embedding model supplying $\text{Rel}(\cdot, \cdot)$) when running the evaluation protocol of Section 5; and (5) report the full hyperparameter set alongside any future results table, given the sensitivity analysis called for in Section 5.3.

5 Evaluation Protocol and Preliminary Results

Because HindsightTag formalizes a mechanism that, to our knowledge, had not previously been implemented or evaluated in any LLM agent memory system, our priority in evaluating it is to specify a complete and falsifiable protocol rather than to lead with headline numbers. Sections 5.1–5.3 give that protocol in full; Section 5.4 then reports preliminary results from a reference implementation run against it on real conversational data, framed explicitly as an initial validation pass rather than a mature, production-scale evaluation. This sequencing follows the precedent of prior neuroscience-grounded memory proposals that were published as architecture and theory contributions accompanied by testable predictions [Lerma-Torres, 2026], while going one step further by executing an initial pass of those predictions ourselves rather than deferring the entire evaluation to future work.

5.1 The Hindsight Rescue Benchmark (HRB) — design

We propose a benchmark constructed specifically to test retroactive rescue behavior, which we are not aware of any existing benchmark [Maharana et al., 2024, Wu et al., 2024, Hu et al., 2025] being designed to measure, since all appear to assume the encoding-time-monotonic assumption of Section 3 in their own ground-truth construction. HRB would consist of four categories of planted scenario, injected into a realistic conversational or document stream:

(a) Rescued precursor:

a deliberately low-salience statement (e.g., an offhand remark) is planted at time τ , followed after a controlled delay $\Delta \in \{1 \text{ hour}, 1 \text{ day}, 1 \text{ week}, 1 \text{ month}\}$ by a high-salience, thematically related trigger event at $\tau + \Delta$. A correctly functioning system should retain and surface the precursor when queried after the trigger, despite its low salience at encoding time.

(b) Heterosynaptic control:

identical to (a), except the trigger event is high-salience but thematically unrelated to the precursor, testing whether content-independent (heterosynaptic) rescue occurs and whether it is desirable.

(c) Negative control (no trigger):

a precursor identical in initial salience to (a) and (b), but with no subsequent high-salience event within W_{capture} , which should decay normally under the host system’s ordinary policy.

(d) Negative control (no precursor):

a high-salience trigger event with no preceding low-salience precursor in its lookback window, verifying that the presence of a capture mechanism does not spuriously alter the host system’s ordinary behavior.

5.2 Metrics

We propose the following metrics, each computable from the four scenario categories above: **Rescue Recall**, the fraction of category (a) precursor-trigger pairs in which the precursor is correctly retained and retrievable at a later query point; **False Rescue Rate**, the fraction of category (c) negative-control precursors that are incorrectly retained, indicating over-eager rescue; **Retention Lift**, the difference in retention probability at a fixed later time between matched precursors in category (a) and category (c); and **Temporal Co-Allocation Recall@k**: for each planted precursor-trigger pair, whether the true trigger ranks among the top- k temporal co-allocates of the precursor against all temporally registered memories in the stream, scored at every planted delay Δ rather than only where $|\tau_{\text{trigger}} - \tau_{\text{precursor}}| \leq E_{\text{window}}$; pairs separated by more than E_{window} cannot form a co-allocation edge and are scored as structural misses, not excluded from the denominator.

5.3 Baselines, ablations, and statistical protocol

We propose comparing HindsightTag attached to a given host system against that same host system alone, using ZenBrain, FadeMem, FSFM, and Mem0 (Section 2) as host systems, since our review found that none of the four implements a rescue mechanism. Based on that review, we predict each should score at or near zero Rescue Recall in its unmodified form; this is a prediction we have not yet tested for ZenBrain, FadeMem, or FSFM specifically, and Section 5.4 reports results only for Mem0 and a vector-store control. We propose ablating ω_{assoc} across $\{0, 0.25, 0.5, 0.75, 1\}$ to characterize the heterosynaptic-to-associative spectrum described in Section 4.4, and sweeping W_{capture} , E_{window} , θ_{capture} , and θ_{rescue} to characterize sensitivity and to guide calibration recommendations for future deployments, seeded from the literature-informed starting ranges given in Section 4.8. To make any future results statistically interpretable, we recommend running each configuration across multiple random seeds and planted-scenario instantiations, reporting bootstrap confidence intervals rather than point estimates alone, and applying a multiple-comparisons correction (e.g., Holm-Bonferroni) when reporting the full hyperparameter sweep described above. As in Section 5.4, we recommend reporting per-arm percentile bootstrap confidence intervals for each individual condition (host alone, host+HindsightTag), and reserving a paired bootstrap on matched (stream, conversation, seed) differences for any claim about an effect *attributable* to HindsightTag itself, such as the non-interference result in Section 5.4.

5.4 Preliminary results

We implemented HindsightTag as described in Section 4.8¹ and ran the HRB protocol of Sections 5.1–5.3 against two host systems — Mem0-style decaying memory (the intended target) and a non-decaying vector store (a control with nothing to rescue) — using 20 real conversations (10 from LoCoMo [Maharana et al., 2024], 10 from LongMemEval [Wu et al., 2024]²) as the background stream, $\times 10$ seeds (200 runs per method), with $\omega_{\text{assoc}} = 0$, $T_{\text{tag}} = 6\text{h}$, $W_{\text{capture}} = 35\text{d}$, $E_{\text{window}} = 4\text{h}$, $\theta_{\text{capture}} = 0.7$, $\theta_{\text{rescue}} = 0.10$ (the literature-informed seeds of Section 4.8). Each per-arm bar in Figure 2 reports a bootstrap 95% confidence interval computed independently on that arm’s 200 runs (percentile method); the attributable HindsightTag effect — e.g., the 0.00 delta reported for the vector-store control — is instead computed via a paired bootstrap on differences matched by stream, conversation, and seed, which is the appropriate comparison for a claim about the plug-in’s effect rather than either arm in isolation. Figure 2 reports all four HRB metrics, stratified by the planted delay Δ .

¹Code and the reference implementation are available at <https://github.com/ViveK1One/hindsighttag>.

²We use the longmemeval-cleaned oracle split rather than the original, now-deprecated longmemeval.s.json release, so that independent replication attempts land on the same underlying file.

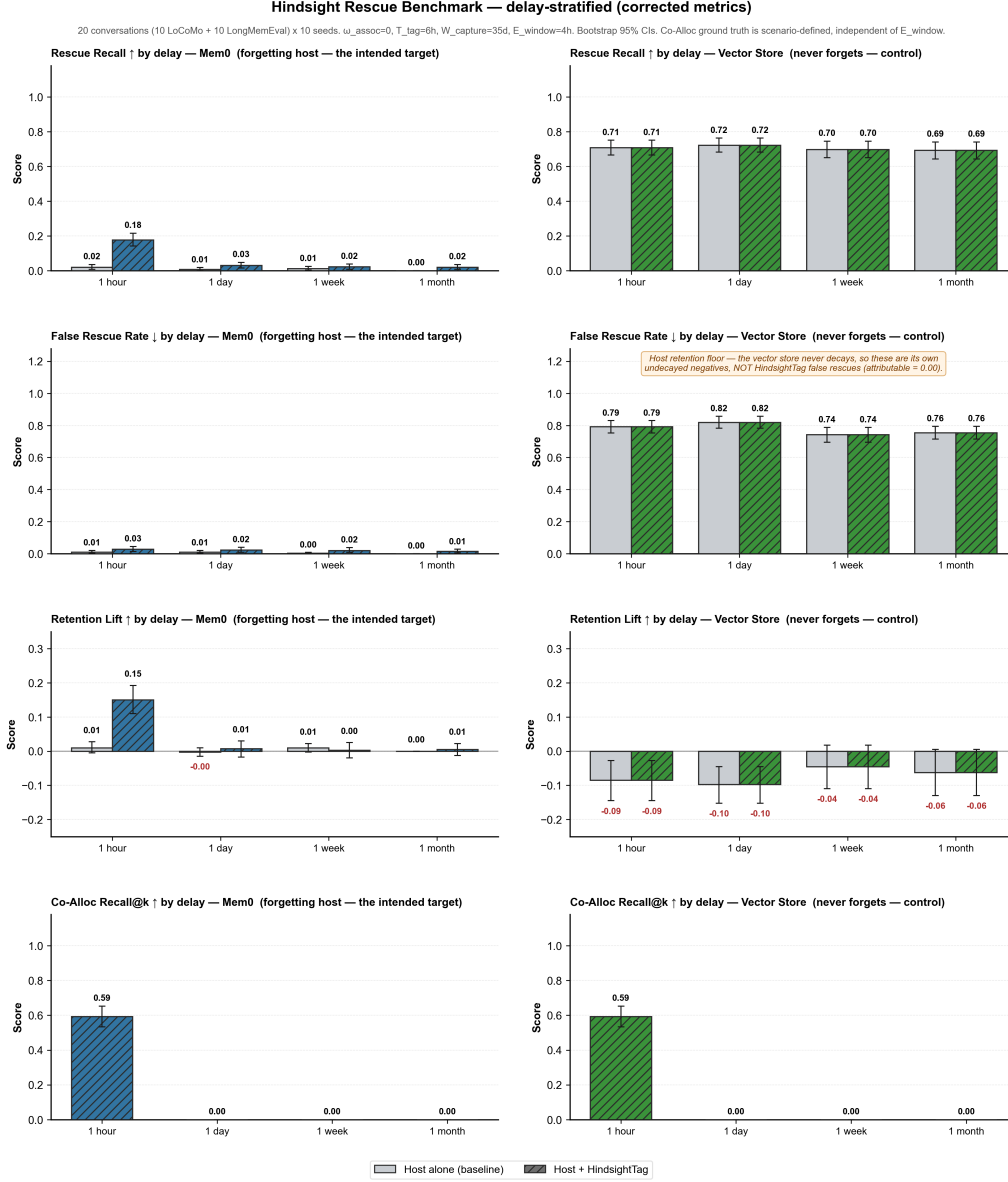


Figure 2: Hindsight Rescue Benchmark results, delay-stratified, for Mem0 (decaying host, intended target) and a vector-store control (never decays). 20 conversations (10 LoCoMo + 10 LongMemEval) $\times$ 10 seeds = 200 runs/method; bars show mean $\pm$ 95% bootstrap CI computed per-arm (percentile method) on that arm’s 200 runs. The Vector Store False Rescue Rate panel reflects that host’s own undecayed retention floor, not an attributable effect of HindsightTag: baseline and +HindsightTag are numerically identical at every delay (attributable contribution = 0.00, via a paired bootstrap on matched differences), which is itself the clearest empirical confirmation of the non-interference property claimed in Section 4.6. Co-Alloc Recall@k is scored at every delay with the same $n = 200$ denominator; beyond 1 hour it is a genuine, well-defined 0.00, not an undefined bucket — the precursor’s true temporal partner lies outside $E_{\text{window}} = 4h$, so no co-allocation edge can form and the pair is a structural miss that is counted, not skipped.

Honest reading of these results. On the intended target host (Mem0-style decay), HindsightTag lifts Rescue Recall from 0.02 to 0.18 at the 1-hour delay (a $9\times$ relative lift) with a correspondingly modest absolute recall, while False Rescue Rate stays low (0.01–0.03) across all four delays; Retention Lift shows the same pattern, peaking at 0.15 at 1 hour and collapsing toward zero beyond it. This decline past 1 hour is not a windowing artifact: $W_{\text{capture}} = 35$ days covers all four tested delays, so nothing is excluded from eligibility by that parameter. It is instead an emergent consequence of $T_{\text{tag}} = 6\text{h}$ — the tag’s capture strength $g_i(t)$ decays to roughly 2% of its initial value by one day, pushing C_i below θ_{rescue} for typical salience values well before W_{capture} would exclude the pair. In other words, the *effective* useful rescue window in this configuration is governed by T_{tag} , not by the nominal W_{capture} bound, and is considerably narrower than W_{capture} alone would suggest; a deployer wanting rescue behavior at longer delays would need to increase T_{tag} , not W_{capture} . On the vector-store control, Rescue Recall, False Rescue Rate, and Retention Lift are numerically identical between baseline and +HindsightTag at every delay, which is exact empirical confirmation, not merely a design intention, of the non-interference property of Section 4.6: with nothing decaying, there is nothing to rescue, and the plug-in changes nothing. Temporal Co-Allocation Recall@k is 0.59 at the 1-hour delay on both hosts — a real measurement against genuine temporal distractors in the surrounding conversation stream, not a tautology, since it requires the true trigger to rank among the top- k temporal neighbors of the precursor rather than merely checking a threshold. Beyond 1 hour it is a genuine, well-defined 0.00, scored with the same $n = 200$ denominator as the 1-hour bucket: the precursor’s true partner lies outside $E_{\text{window}} = 4\text{h}$, so no co-allocation edge can form and the pair is a structural miss that is counted, not skipped. These by-construction misses are included when the metric is aggregated across delays (aggregate Co-Alloc Recall@k = 0.148, i.e., 0.59 at 1 hour averaged with genuine zeros at the three longer delays), so the drop-off is a direct, disclosed consequence of E_{window} — in contrast to the Rescue Recall drop-off above, which is an emergent consequence of T_{tag} rather than a parameter boundary.

5.5 Scope and evaluation status

We emphasize three scope limitations on the results above. First, they come from a single reference implementation and two conversational data sources (LoCoMo, LongMemEval); we have not yet run the full hyperparameter sweep, host-system set (ZenBrain, FadeMem, FSFM), or ω_{assoc} ablation described in Section 5.3, and the six hyperparameters used here remain the literature-informed calibration seeds of Section 4.8, not tuned values. Second, the headline lift is carried almost entirely by the 1-hour stratum; absolute Rescue Recall even in that best case is 0.18, not a large number in isolation, and we report it alongside the relative lift rather than in place of it. Third, independent replication has not yet occurred. We therefore present Section 5.4 as an initial, limited-scale validation confirming the mechanism behaves as specified — rescuing precursors within its effective window, leaving non-decaying hosts untouched, and supporting genuine temporal-neighbor retrieval — rather than as a mature claim of empirical superiority over any specific competing system. Any such comparative claim must await the fuller sweep described in Section 5.3, ideally including independent replication.

6 Discussion

6.1 Relationship to reconsolidation

HindsightTag’s rescue mechanism should be clearly distinguished from memory reconsolidation [Nader et al., 2000] and from ZenBrain’s implemented ReconsolidationEngine [Bering, 2026], both of which describe a memory becoming labile and editable upon its *own* retrieval. HindsightTag’s rescue operator is triggered by the arrival of a *different*, later event and acts on memories that are not currently being retrieved at all. The two mechanisms are complementary rather than competing: a host system’s reconsolidation

engine governs what happens when a memory is deliberately revisited, while HindsightTag governs what happens to memories that are never revisited but whose significance is later revealed by unrelated events.

6.2 Governance and auditability

Any mechanism that can autonomously change a memory’s retention status without an explicit user action raises governance questions. We consider the provenance record specified in Section 4.8 — which permanently logs the triggering event, timestamp, and computed capture strength for every rescue — a non-optional component of the design rather than an implementation detail, since it allows a human reviewer or downstream audit process to reconstruct exactly why any given memory was retained. We recommend that any production deployment of HindsightTag expose this log directly to end users or administrators, particularly in settings involving personal or organizational data.

6.3 A motivating application: organizational memory

The mechanism described here is motivated in part by a recurring failure pattern in organizational memory systems that ingest heterogeneous communication streams (chat, email, meeting notes, and structured records). A casual, low-salience remark — for instance, an early, informal suggestion about a future pricing change — is typically indistinguishable at encoding time from thousands of similarly low-salience messages, and is therefore discarded by any encoding-time-monotonic system long before it might later prove to be the origin of a significant decision. A system implementing HindsightTag would, upon the later high-salience event (the actual pricing change being enacted), retroactively identify and surface the earlier remark as precedent, a capability that Section 2 confirms is absent from all systems examined in this study, including those designed explicitly for organizational and enterprise memory use cases.

7 Limitations

We list the limitations of this work candidly, in the tradition of the neuroscience-grounded architecture papers it builds on. First and most importantly, the empirical evaluation in Section 5.4 is preliminary and limited in scale: a single reference implementation, two conversational data sources, no independent replication yet, and a headline lift carried almost entirely by the shortest tested delay. It confirms the mechanism behaves as formally specified in Section 4 — non-interference on a non-decaying host, window-bounded rescue on a decaying one — but does not constitute a mature, production-scale validation, and the full baseline/ablation sweep of Section 5.3 remains future work. Second, the mechanism introduces five new hyperparameters (T_{tag} , W_{capture} , θ_{capture} , θ_{rescue} , ω_{assoc}) and a sixth for the companion mechanism (E_{window}); the literature-informed starting ranges given in Section 4.8 are calibration seeds, not validated defaults, and inappropriate settings could plausibly produce either negligible rescue behavior or, at the opposite extreme, excessive false-rescue noise that degrades rather than improves memory quality. Third, HindsightTag’s rescue quality is bounded by the host system’s own salience function $\text{Sal}(\cdot)$; it adds a new temporal mechanism but does not improve the underlying judgment of what counts as salient. Fourth, we have not evaluated interaction effects between the rescue mechanism and a host system’s own consolidation or reconsolidation pathways, which could plausibly compound or interfere with one another in ways not captured by the additive design of Section 4.6. Fifth, this paper addresses a single-agent memory setting; extending temporal co-allocation and rescue across multiple agents or users sharing a memory store, with attendant privacy and access-control considerations, is left entirely to future work. Sixth, our verification of absence in prior work is uneven in strength: for ZenBrain and the theoretical architecture of Lerma-Torres [2026] (Section 1), we verified the relevant claims by direct quotation of the source text; for the remaining systems discussed in Section 2, including MIRIX, our claims rest on published technical descriptions rather than independent

code inspection, and we did not search patents or non-indexed industry documentation. Commercial systems (Supermemory, Zep, Mem0) in particular may implement undocumented behavior that differs from their public reports. Relatedly, the claim in Section 5.3 that ZenBrain, FadeMem, and FFSM would score near-zero Rescue Recall in unmodified form is a prediction based on their published mechanisms, not a measured result; only Mem0 and a vector-store control were actually run (Section 5.4). Seventh, we have not empirically compared temporal co-allocation (Section 4.5) against a simple session- or time-bucket baseline drawn from the working-set/temporal-locality caching tradition, so we cannot yet quantify what the specific formalization in that section adds over that simpler alternative. Finally, our claim that this specific mechanism is unclaimed in the existing literature is based on a thorough but necessarily time-bounded search of the arXiv and neuroscience literature as of mid-2026, scoped to published work as detailed above; given the pace of publication in this area, we recommend that any subsequent submission of this work be preceded by a fresh literature check immediately before submission.

8 Conclusion

We have identified a specific, verifiable structural gap — the encoding-time-monotonic salience assumption — shared by the surveyed literature on LLM agent memory, including its two most recent and most explicitly neuroscience-grounded members. We have formalized, to the best of our knowledge for the first time at the level of discrete symbolic memory records, the synaptic tagging-and-capture and behavioral tagging phenomena from the memory neuroscience literature as a bounded, auditable, non-interfering plug-in mechanism, together with a companion mechanism for content-independent temporal co-allocation drawn from the engram-allocation literature. Because this specific discrete, threshold-gated, auditable backward-rescue plug-in is, to our knowledge, unimplemented elsewhere — notwithstanding concurrent synaptic-inspired work on continuous retrieval scoring such as SynapticRAG [Hou et al., 2024] — we have accompanied it with a reference implementation, a complete evaluation protocol, and preliminary results (Section 5.4) that empirically confirm its non-interference property and reveal a window-bounded rescue effect governed by the tag-decay constant T_{tag} rather than the nominal capture window W_{capture} . These results are an initial, limited-scale validation, not a mature production claim; we view scaling the evaluation to the full baseline and ablation sweep of Section 5.3, and securing independent replication, as the immediate and necessary next steps for this line of work.

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A Algorithm (pseudocode)

Algorithm 1 below gives the HindsightTag capture-and-rescue cycle. Capture detection and the backward rescue transition (the `for each new event e` block) execute on `store(e)`, immediately as each event arrives; tag decay and expiry (the final loop) run on the host system’s periodic `decay()/tick()` operation. Both use the interface contract of Section 4.8.

Algorithm 1. HindsightTag capture-and-rescue cycle.

```

Input: memory store M, temporal graph T, current time t
Input: hyperparameters T_tag, W_capture, theta_capture,
      theta_rescue, omega_assoc, E_window
Requires: host interface get_salience, get_timestamp,
          get_relatedness, get_consolidation_threshold
for each new event e arriving at time t:
    if get_salience(e) < get_consolidation_threshold():
        initialize tag g_i(t) for the newly written m_i [Eq. 2]
        register m_i in T with eligibility window E_window,
        linking to all m_j with |tau_i - tau_j| <=
            E_window [Eq. 6]
    if get_salience(e) >= theta_capture: # capture event
        T_active = { m_i in M : tau_i in
                    [t - W_capture, t), g_i(t) > 0 } [Eq. 3]

```

```

for each m_i in T_active:
    C_i = g_i(t) * get_salience(e) *
        (omega_assoc * get_relatedness(e, m_i)
        + (1 - omega_assoc))
    if C_i >= theta_rescue:
        v_i(t) <- min(1, v_i(t) + Delta_rescue *
            (1 - v_i(t)) * C_i)
        mark m_i as consolidated
        append {rescued_by: e, at: t,
            capture_strength: C_i,
            omega_assoc_used: omega_assoc,
            mode: (heterosynaptic if
                omega_assoc == 0 else
                associative)}
            to m_i.provenance
        # tie-breaking: if multiple m_i reach C_i >=
        # theta_rescue simultaneously, process in
        # increasing memory_id order (Sec. 4.8)
for each tagged, not-yet-consolidated m_i:
    update g_i(t)
    if g_i(t) == 0:
        drop tag (no effect on v_i); log expiry per the
        tombstone record format of Sec. 4.8

```

[Eq. 4]

[Eq. 5]

[Eq. 2]